

transparency on the SIP and force more participants to purchase costly direct feeds to maintain the same level of transparency of liquidity.”⁷⁹⁴

Although the Commission agrees that both the round lot definition adopted in the MDI Adopting Release and the amended minimum pricing increments will impact the NBBO and will result in a narrower spread for impacted NMS stocks, the NMS stocks that would be subject to both the round lot definition and the amended minimum pricing increments are likely to be very small in number and also extremely liquid, which could counteract any potential harm to liquidity resulting from the interaction of both changes. The round lot definition adopted in the MDI Adopting Release and the amended minimum pricing increments each will impact the NBBO and each will result in a narrower spread for those NMS stocks that are assigned a smaller round lot⁷⁹⁵ or a smaller minimum pricing increment.⁷⁹⁶

The round lot definition will narrow the spread for NMS stocks that have an average closing price over \$250 per share by showing better prices for these stocks. The amended minimum pricing increments will reduce the spread for those NMS stocks that have a narrow TWAQS and will allow these stocks to be priced more competitively in smaller increments, which will more accurately reflect supply and demand. Accordingly, the smaller round lot and the smaller minimum pricing increment narrow spreads in different ways. In response to the comment stating that the round lot definition and the proposed tick size changes are duplicative

⁷⁹⁴ See Virtu Letter II at 10.

⁷⁹⁵ See infra section VII.D.4.a.

⁷⁹⁶ See infra section VII.B.2; section VII.D.1.

because they would both result in narrow spreads,⁷⁹⁷ both requirements will narrow spreads, but they are not duplicative.

Although there may be NMS stocks that are assigned both a smaller round lot and a smaller minimum pricing increment, Commission analysis of data discussed below shows that this overlapping universe of NMS stocks is very small. In other words, most NMS stocks will not be assigned both a round lot that is less than 100 shares and a smaller \$0.005 minimum pricing increment and therefore will not experience a combined impact on the NBBO spread or depth. As explained in the analysis, since only a few NMS stocks are expected to be subject to both a smaller round lot and a smaller tick size, the potential combined impact of the amendments to the minimum pricing increments and round lots should be limited.⁷⁹⁸ Specifically, in response to comments expressing concerns about the combined impact of the proposed smaller minimum pricing increments and the implementation of the round lot definition,⁷⁹⁹ the Commission conducted the analysis to determine the magnitude of NMS stocks that would be impacted by both changes. According to the Commission's analysis, as of November 30, 2023, only 163 NMS stocks were priced above \$250.00 per share and would have been potentially eligible to be assigned to a round lot size smaller than 100 shares, out of a universe of 11,200 NMS stocks on that date.⁸⁰⁰ Further, based on Commission review of the 163 NMS stocks that would have been assigned to a round lot less than 100 shares, as of November 30, 2023, only two out of the 163

⁷⁹⁷ See ASA Letter at 6.

⁷⁹⁸ See *infra* section VII.D.4.a. (discussing the impact of the acceleration of the implementation of the round lot definition and reduced tick sizes and stating that the number of stocks trading over \$250 with spreads narrower than \$0.015 is "likely very small").

⁷⁹⁹ See, e.g., Virtu Letter II at 10; Citadel Letter I at 26; ASA Letter at 6.

⁸⁰⁰ The Proposing Release stated that, based on average closing prices on the primary listing exchange in Mar. 2022, there were 181 NMS stocks priced over \$250. See Proposing Release, *supra* note 11, at 80300 n.407.

had an average quoted spread over the previous thirty trading days of \$0.015 or less and therefore may have been potentially eligible to have been assigned to the smaller \$0.005 minimum pricing increment.⁸⁰¹ The two NMS stocks—SPY and QQQ—are among the most liquid exchange-traded products.⁸⁰² For the month of November 2023, SPY had the highest average daily traded value, while QQQ was ranked second in average daily traded value.⁸⁰³

Based on this information, while the Commission recognizes that the interaction of the minimum pricing increment and the round lot definition may result in some reduction of depth for the very few NMS stocks that may be subject to both a smaller tick size and a round lot size of less than 100 shares,⁸⁰⁴ the impact of the reduction of NBBO depth should not be of such a level as to impede trading in the affected NMS stocks because these stocks are highly liquid, which should greatly mitigate the impact of reduced depth at the NBBO.⁸⁰⁵ Furthermore, these changes will benefit investors and other market participants trading these stocks through more accurate pricing and a reduction in spreads. While the extent of any reduced depth at the NBBO is not known at this time, due to the volume traded in these two NMS stocks, any potential

⁸⁰¹ These NMS stocks were ETFs: SPY and QQQ. As of Nov. 30, 2023, the last sale price for SPY was \$454.30 and its average bid-ask spread over the previous 30 trading days was \$0.0105. For QQQ, as of Nov. 30, 2023, the last sale price was \$386.70 and the average bid-ask spread over the previous 30 trading days was \$0.0116. This analysis was conducted using Bloomberg data. See also supra section III for a discussion of the amended minimum pricing increments. The calculation of a TWAQS for NMS stocks will occur during an Evaluation Period for purposes of assigning minimum pricing increments. See Rule 612(a)(1).

⁸⁰² SPY and QQQ are also among the most liquid NMS stocks. Using Bloomberg data, for the period Jan. 22, 2024 – Feb. 16, 2024, SPY had the highest average daily traded value of all NMS stocks, while QQQ was ranked fourth. Specifically, for this period, the average daily traded value per day for SPY was \$36,581,363,712, or 5.9% of total value traded of all U.S. equity trading, and for QQQ the average daily traded value per day was \$18,632,960,000, or 3.5% of total value traded of all U.S. equity trading.

⁸⁰³ Based on daily average traded value for Nov. 1, 2023 – Nov. 30, 2023, using Bloomberg data. For SPY, the daily average traded value for Nov. 2023 was \$31,652,396,337. For QQQ, it was \$17,527,314,000.

⁸⁰⁴ See infra section VII.D.4.a.

⁸⁰⁵ See infra section VII.D.4.a. (stating, “[t]he exceptional liquidity of the affected stocks will likely protect their NBBO from material deterioration.”).

reduction will not impair market participants' ability to trade these stocks because these stocks would be among the most liquid and therefore easily traded.⁸⁰⁶

Additionally, some liquidity that is consolidated at the preexisting round lot⁸⁰⁷ and \$0.01 minimum pricing increment may be reflected in the adopted round lot sizes and smaller minimum pricing increment. Specifically, some odd-lot orders are currently aggregated into the 100-share round lot.⁸⁰⁸ Upon implementation of the round lot definition, some orders that were considered odd-lots may be of round lot size as defined. Further, interest that is displayed at the previously required \$0.01 minimum pricing increment may be reflected in orders that are entered in the smaller tick size. Once these amendments are implemented, the NBBO will reflect better prices, both because of the smaller round lot size for some NMS stocks and new \$0.005 increment for some other NMS stocks. As smaller sized orders in higher priced stocks are often priced better than orders that are currently in round lots, the smaller round lot sizes will allow potentially better priced orders to be the basis of the NBBO.⁸⁰⁹ The new \$0.005 increment will also result in the NBBO reflecting better prices because the smaller increment will allow orders to be priced in a manner that is more reflective of the supply and demand of liquidity for the stock.⁸¹⁰ Accordingly, each of these amendments will result in narrower NBBO spreads and better prices.⁸¹¹ Further, those market participants that may need to trade in large sizes may be

⁸⁰⁶ Id.

⁸⁰⁷ See supra note 66.

⁸⁰⁸ See MDI Proposing Release, supra note 744, at 16738-39 (describing exchange rules on aggregating odd-lot across multiple prices and providing them to the exclusive SIPs at the least aggressive price if the combined odd-lot interest is equal to or greater than a round lot).

⁸⁰⁹ See Proposing Release, supra note 11, at 80294; MDI Adopting Release, supra note 10, at 18616.

⁸¹⁰ See infra section VII.D.1; section VII.D.1.b.i; section VII.D.1.b.ii; section VII.E.3.

⁸¹¹ See infra section VII.D.1 and note 1145 and accompanying text.

able to see liquidity outside of the NBBO by considering the new odd-lot information that will be available in SIP data as well as depth of book data that is available via exchange proprietary data feeds.⁸¹²

In response to the comment that raised concerns about the potential for the proposed variable minimum pricing increments and the new round lots to reduce transparency on the exclusive SIPs,⁸¹³ for the reasons discussed above,⁸¹⁴ the combined impact of the adopted minimum pricing increments and round lot definition should not reduce transparency for most NMS stocks and the exclusive SIPs will provide a more accurate NBBO once these amendments are implemented. While the round lot will be smaller for certain NMS stocks, as described above, the NBBO based on the new round lots will in many cases reflect better prices.⁸¹⁵ Therefore, while the actual number of shares will be smaller for certain NMS stocks, the disseminated prices will likely be better.⁸¹⁶ In addition, as discussed above, the new minimum pricing increment required under Rule 612 for certain NMS stocks will allow the NBBO that is disseminated by the exclusive SIPs to reflect more competitive pricing. These amendments will

⁸¹² Once implemented, the MDI Rules will add depth of book information to consolidated market data, and this information will provide information about depth outside of the NBBO for those market participants that would find this information useful. See infra section VII.D.4.a.; MDI Adopting Release, supra note 10, at 18728, 18730 (explaining that SIP data currently only includes top of book quotes). 17 CFR 242.600(b)(26)(I) (defining “core data” to include depth of book data) and 17 CFR 242.600(b)(24)(i) (defining “consolidated market data” to include core data).

⁸¹³ See Virtu Letter II at 10.

⁸¹⁴ See supra notes 807-812 and accompanying text.

⁸¹⁵ See supra section V.B.1. (Table 1 and Table 2).

⁸¹⁶ See MDI Adopting Release, supra note 10, at 18742, 18743.

enhance the NBBO that is calculated and disseminated by the exclusive SIPs by reflecting more competitive and better available prices.⁸¹⁷

Commenters also expressed concern that the implementation of the round lot definition and the proposed changes to the minimum pricing increments would confuse investors.⁸¹⁸ One commenter warned that changes to round lots and tick sizes would confuse retail investors and reduce trust in the market.⁸¹⁹

Because there are expected to be only a small number of NMS stocks that could be subject to both a change in a minimum pricing increment and a change to the round lot size, the risk of investor confusion is limited. Market participants may choose to educate investors about the new round lot and amended minimum pricing increments, as they sometimes choose to educate investors regarding other regulatory changes that impact how investors enter orders. Investors are already familiar with three round lot sizes⁸²⁰ and two minimum pricing increments,⁸²¹ so the addition of only one round lot size and one minimum pricing increment is unlikely cause investor confusion.⁸²² The Commission is also adopting new indicators for dissemination on the exclusive SIPs of the assigned round lots and minimum pricing increments to alert market participants, including investors, of the relevant round lot and minimum pricing

⁸¹⁷ See supra section III.C. See also MDI Adopting Release, supra note 10, at 18744, 18745. See also infra section VII.D.4.a. (stating that the round lot definition would shrink the NBBO for stocks priced greater than \$250, would increase transparency and would result in better order execution).

⁸¹⁸ See, e.g., Tastytrade Letter at 22; SIFMA Letter II at 34; Morgan Stanley Letter at 3.

⁸¹⁹ See, e.g., Tastytrade Letter at 22.

⁸²⁰ Under exchange rules, there are three different round lot sizes. See supra note 67. The MDI Rules' round lot definition adds one more round lot size, i.e., 40 shares. Consistent with its views stated here, the Commission previously considered potential investor confusion with the additional round lot size and did not believe it will be confusing to investors. See MDI Adopting Release, supra note 10, at 18618.

⁸²¹ Preexisting Rule 612 included two minimum pricing increments based on the price of a quote or order - \$0.01 and \$0.0001.

⁸²² See supra section III. See also infra notes 1589-1593 and accompanying text.

increment for each NMS stock. These indicators will also help to mitigate concerns about any potential for investor confusion.⁸²³

ii. Comments on the Round Lot Indicator

The Commission proposed to amend Rule 600(b)(78) to add a requirement to make the indicator of the applicable round lot size available to the exclusive SIPs in Rule 600(b)(78)(iv).⁸²⁴ The Commission is adopting this requirement as proposed in Rule 600(b)(89)(iv).⁸²⁵ The Commission did not receive comments specifically supporting or objecting to the proposed amendment. However, two commenters cited this requirement as support for their arguments that the proposed 90-day compliance deadline for the round lot and odd-lot information definitions would provide an insufficient amount of time.⁸²⁶ As discussed below,⁸²⁷ the Commission is providing more time for compliance with the round lot definition, which is a substantially longer period for compliance than the 90 days that was proposed.⁸²⁸ The Commission is providing a longer compliance period than proposed after considering the information provided by commenters that requested more time to comply with the implementation of the round lot definition, including implementation of the round lot indicator.⁸²⁹ The additional time will provide market participants with time to make the changes necessary to implement the round lot indicator.

⁸²³ See infra note 1385; MDI Adopting Release, supra note 10, at 18619.

⁸²⁴ See Proposing Release, supra note 11, at 80299.

⁸²⁵ See supra note 320.

⁸²⁶ See NYSE Letter I at 7-8; Letter from Robert Books, Chair of the Operating Committee, Operating Committees of the CTA Plan, CQ Plan and UTP Plan, dated Mar. 28, 2023 (“CTA, CQ, UTP Plans Operating Committees Letter”) at 3. See also infra section VI.C.

⁸²⁷ See infra section VI.C.

⁸²⁸ See Proposing Release, supra note 11, at 80300-01.

⁸²⁹ See supra note 826; infra section VI.C.

iii. Comments on the Round Lot Definition

The Commission received comments on the round lot definition that was adopted in the MDI Rules.⁸³⁰ Commenters raised concerns about the defined round lot sizes,⁸³¹ the determination of round lot size based on price,⁸³² the impact of smaller round lot sizes on the relevance of the NBBO,⁸³³ and the impact of the round lot definition as well as the odd-lot information definition on bandwidth.⁸³⁴ The Commission considered and addressed issues related to adopting the round lot definition and the odd-lot information definition in the MDI Adopting Release.⁸³⁵

One commenter stated that investors trading in options may be confused by the round lot definition, stating that retail investors who trade options know one options contract represents 100 shares or a “round lot.”⁸³⁶ The Commission considered and addressed the interaction of the new round lot definition and options trading in the MDI Adopting Release.⁸³⁷ Further, it is unlikely that the acceleration of the round lot definition could confuse retail investors trading in options. Specifically, the round lot size will not change the size of the options contract and precedent exists for standard options contracts on stocks with a round lot size less than 100

⁸³⁰ See, e.g., RBC Letter at 5; Tastytrade Letter at 21; T. Rowe Price Letter at 4; Pragma Letter at 8-9; Data Boiler Letter I at 8 and Data Boiler Letter II at 2.

⁸³¹ See, e.g., Pragma Letter at 1, 8, 9; T. Rowe Price Letter at 4.

⁸³² See, e.g., Tastytrade Letter at 21.

⁸³³ See, e.g., RBC Letter at 5.

⁸³⁴ See Data Boiler Letter I at 8; see also Data Boiler Letter II at 2. The commenter raised concerns about the round lot and odd-lot information definitions.

⁸³⁵ See MDI Adopting Release, supra note 10, at 18615-22.

⁸³⁶ See Tastytrade Letter at 22.

⁸³⁷ See MDI Adopting Release, supra note 10, at 18619, 18747.

shares.⁸³⁸ Furthermore, corporate actions, such as rights offerings, stock dividends, and mergers can result in adjusted contracts representing stock in amounts other than 100 shares, so investors have some familiarity already with options on underlying NMS stocks that have a “round lot” that is less than 100 shares.⁸³⁹

Several commenters also suggested eliminating the concept of round lots altogether.⁸⁴⁰ The Commission is not eliminating the concept of round lots. As the Commission stated in the MDI Adopting Release, round lot orders continue to play an important role in the national market system by delineating orders of meaningful size and focusing regulatory requirements and protections—such as those set forth in rules 602, 604 and 611 of Regulation NMS—on such orders.⁸⁴¹ Further, as the Commission stated in the MDI Adopting Release, eliminating the concept of a round lot could also cause investor confusion and other unintended consequences.⁸⁴²

iv. Modified Round Lot Assignment Frequency and Evaluation Period for Round Lots

Under the MDI Rules, each NMS stock was assigned a round lot size every month based on its average closing price for the prior calendar month on its primary listing exchange.⁸⁴³ Commenters raised concerns about potential confusion and operational risks arising due to the fact that round lots and the proposed minimum pricing increments would be changed at different

⁸³⁸ For example, as of Oct. 26, 2023, one NMS stock has a round lot size of 10 shares while also possessing an option contract size of 100 shares.

⁸³⁹ See MDI Adopting Release, supra note 10, at 18619. See also infra section VII.D.4.a.

⁸⁴⁰ See, e.g., Nasdaq Letter I at 34-35; XTX Letter at 5; Angel Letter at 2-3; Anonymous Letter (Feb. 12, 2023) (stating that order sizes should be treated the same, regardless of status as a round lot or an odd-lot).

⁸⁴¹ See MDI Adopting Release, supra note 10, at 18618 n.274.

⁸⁴² See MDI Adopting Release, supra note 10, at 18618 n.274.

⁸⁴³ MDI Adopting Release, supra note 10, at 18617.

times,⁸⁴⁴ and several commenters suggested aligning the assignment of round lots and minimum pricing increments, either on a quarterly or on a semiannual basis.⁸⁴⁵ One commenter stated “having two to four adjustments per year strikes the appropriate balance between having the optimal round lot and minimum pricing increment with reducing the time that market participants are adjusting to the changes.”⁸⁴⁶ Another commenter warned that having differing assignment schedules for round lot sizes and minimum pricing increments “will materially elevate systemic risk since it only requires a single large market participant to create widespread disruption by failing to properly modify their systems.”⁸⁴⁷ The commenter suggested reducing the frequency of changes to quarterly or semiannually and synchronizing the round lot and minimum pricing increment changes.⁸⁴⁸

After further consideration, the Commission agrees with the concerns raised by the commenters. Frequent systems changes and updates can introduce risks for market participants and the market, and frequent changes to the terms of how an order is entered for an NMS stock can potentially cause investor confusion. Therefore, the Commission is amending the round lot

⁸⁴⁴ See, e.g., FIF Letter at 13; Hudson River Letter at 2, 4; BlackRock Letter at 9, 10. See also SIFMA Letter II at 34, 35 (stating that “The Tick Size Proposal would make dynamic three components of trading that are static today: (i) tick sizes; (ii) access fees; and (iii) round lots. Exacerbating this complication, tick sizes would adjust quarterly, while round lots would change monthly. Tick sizes would be based on average quoted spread, while round lots are based on a stock’s price.”); Morgan Stanley Letter at 3 (stating that the proposed amendments would require “market participants to make frequent changes to their systems...” that “...the risk of technology failure created by monthly and/or quarterly changes to systems by the buy-side, sell-side, exchanges and vendors (including security information processors) may introduce new market and operational risks...” and that “...this dynamic aspect of the proposal could create investor confusion and potentially drive trading inefficiencies.”).

⁸⁴⁵ Two commenters suggested aligning the assignment of round lots and minimum pricing increments on a quarterly or semiannual basis. See Hudson River Letter at 2, 4; BlackRock Letter at 10. One commenter suggested aligning both assignments on a quarterly basis. See FIF Letter at 13. See also *infra* notes 849-853 and accompanying text.

⁸⁴⁶ Hudson River Letter at 4.

⁸⁴⁷ BlackRock Letter at 10.

⁸⁴⁸ See *id.*